INSTITUT FOR MATEMATIK
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Statistical Models

Week 12: Linear models, glms, and "penalized" regression in a Bayesian setting

comments

This week's main topic is the practical analysis of linear models in a Bayesian way, starting with normal distributed models.

For normal linear regression models, we will notice interesting facts about priors: some priors have "shrinkage effects" which means that they implicitely perform variable selection. To understand this, we will shortly look at ridge regression and LASSO in the frequentist world —some of you might know these methods from other courses.

Finally, we will discuss Bayesian analysis of generalized linear models, conveniently using the JAGS machinery. For binary data, we can set up a particular Gibbs algorithm that incorporates a latent variable — you might recall this type of modelling from the first part of the course.

Relevant Readings

- ☐ Analysis of normal means: Reich & Ghosh, Section 4.1
- ☐ Linear models and variable selection: Reich & Ghosh, Section 4.2
- ☐ Generalized linear models: Reich & Ghosh, Section 4.3

Lecture Material

- Link to onenotes on Google drive
- Toy example ridge and LASSO regression

Exercises

tba

Aufgabe 21 zuletzt, som ugens nød